

# As We May Think Next:

Immersive Sensemaking from the Memex to the Chatbot

2026-10-03

**Mario Romero**

Senior Associate Professor  
Immersive Visualization

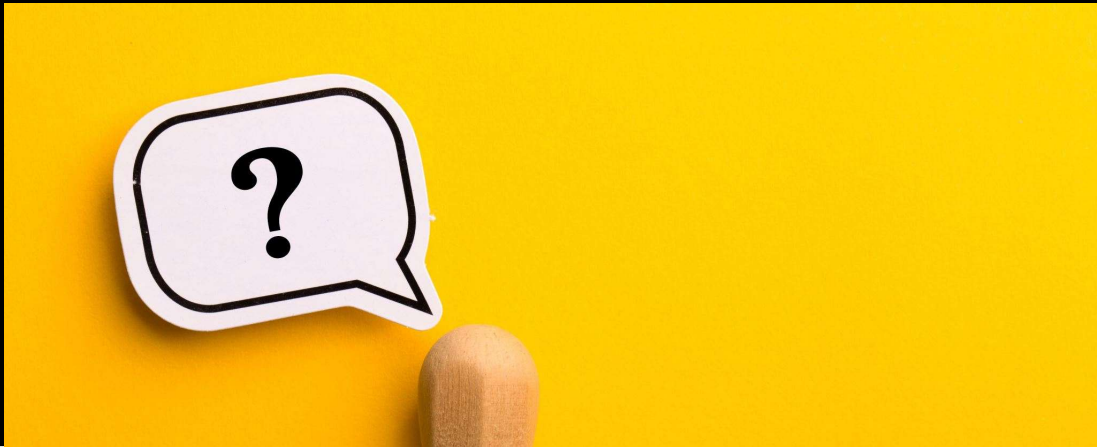


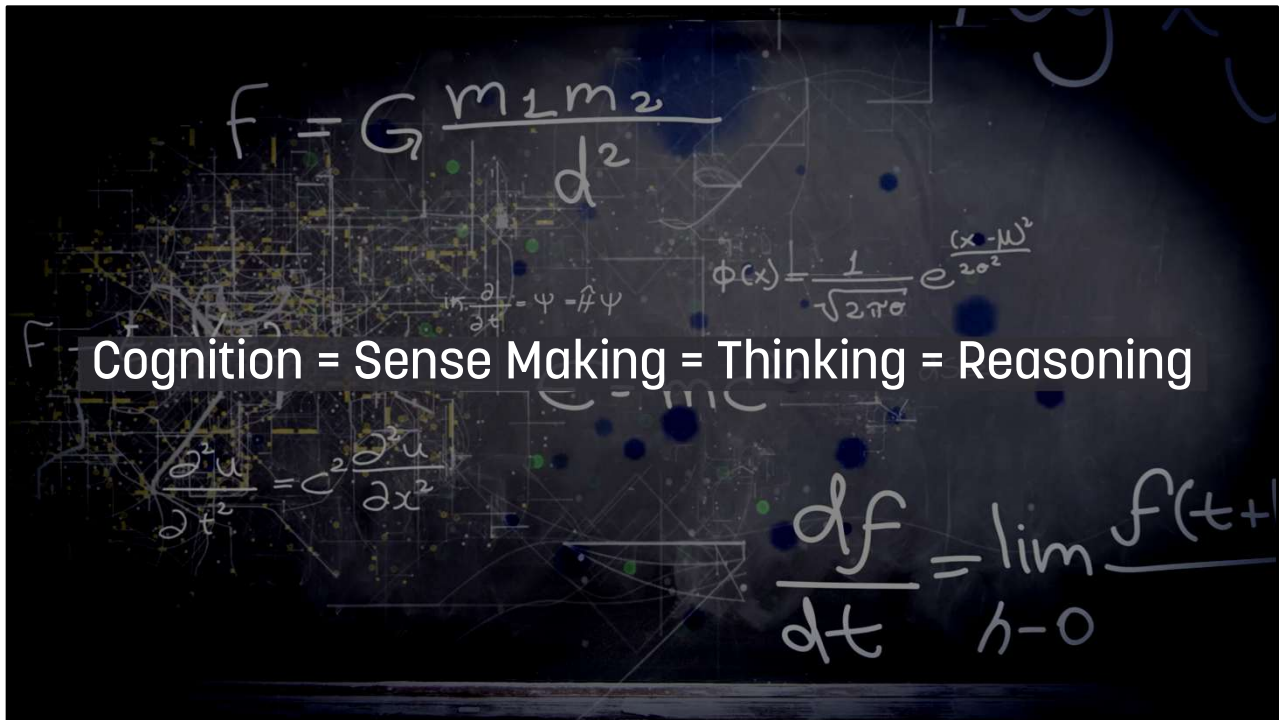
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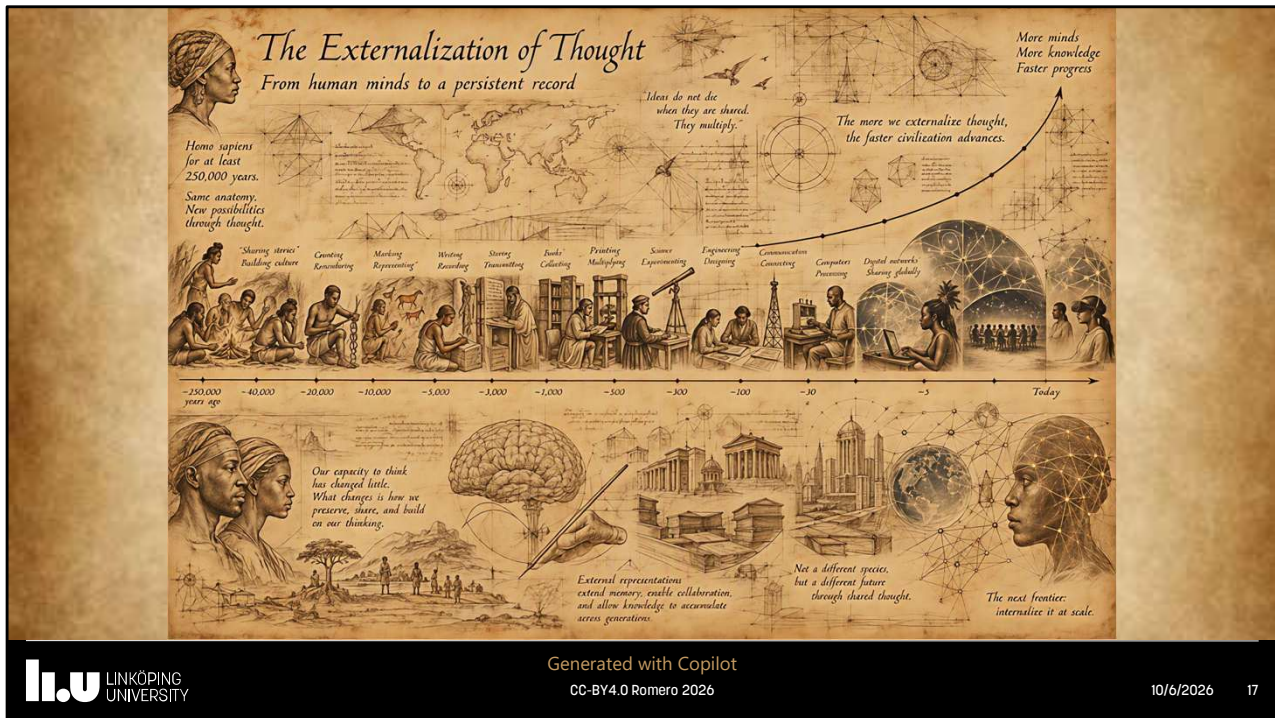
Hi, my name is Mario Romero, senior associate professor of immersive visualization at Linköping university.

# How should we use AI for immersive cognition?





Cognition = Sense Making = Thinking = Reasoning



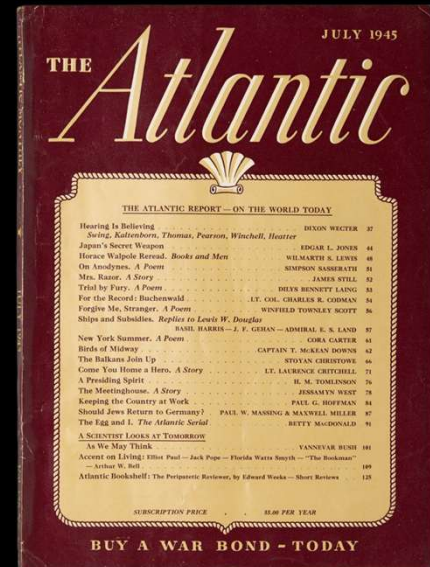
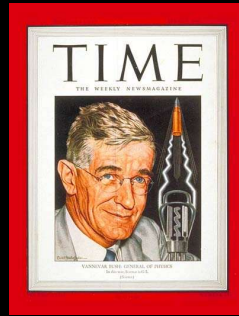
Let us now zoom out. The game annotations are a tiny example of a much larger human pattern. Civilization advances by externalizing thought. Language lets one mind affect another. Marks and knots preserve quantities and events. Writing makes statements durable. Diagrams preserve relationships that are difficult to hold in memory. Books allow ideas to travel across distance and outlive their authors. Scientific records make observations inspectable, contestable, and cumulative. Digital networks increase the speed, reach, and recombination of these externalized artifacts. This does not mean that thought began with writing. Humans were deeply immersed in thought long before formal notation. The transformation is that externalization reduces brittleness. An idea no longer must disappear when the thinker forgets it or dies. Someone else can recover it, misunderstand it, correct it, or build upon it. External cognition therefore supports distributed cognition across people, tools, places, and generations. The crucial question for this talk is what happens when these external and distributed resources become so immediate and responsive that they begin to feel like part of a continuous internal process.

## The Memex

Vannevar Bush

1945

As we may think



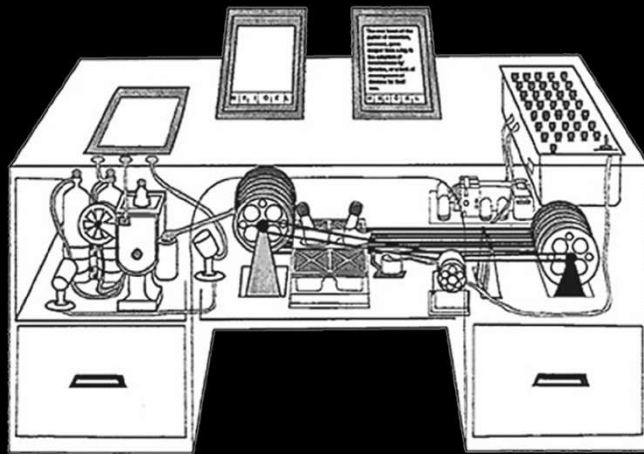
In 1945, Vannevar Bush published *As We May Think* and described the Memex. It was a hypothetical personal device for storing and consulting books, records, and communications. Its lasting idea was not merely storage. Bush emphasized associative trails. Human thought often moves by association rather than by alphabetical categories, so the user should be able to connect items, return to those connections, annotate them, and share the trail.

# Sensemaking

Vannevar Bush

1945

Memory Index - Memex



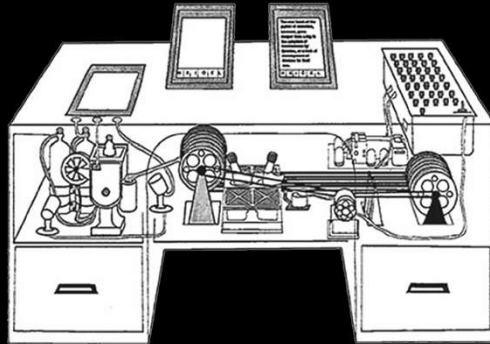
The Memex was not artificial intelligence. It was a proposal for making externalized knowledge more personal, navigable, and compatible with human association. Bush was responding to an information problem that still sounds familiar: knowledge was accumulating faster than researchers could absorb and connect it. The Memex promised an enlarged supplement to memory. For my argument, it is also a bridge. It places a huge external record close to the individual thinker, but the person still navigates the archive through explicit operations. Select, retrieve, connect, annotate, and return. The external record becomes more intimate, yet it remains visibly outside the thinker as a system of documents and trails.

[Sources]

- <https://www.theatlantic.com/magazine/archive/1945/07/as-we-may-think/303881/>
- <https://academic.oup.com/mind/article/LIX/236/433/986238>

[/Sources]

# Sensemaking = Association



# Machine Intelligence

Alan Turing

1950

Computing Machinery and Intelligence



MSS. and other Communications for the Editor should be addressed to  
Prof. G. RYLE, Magdalen College, Oxford.

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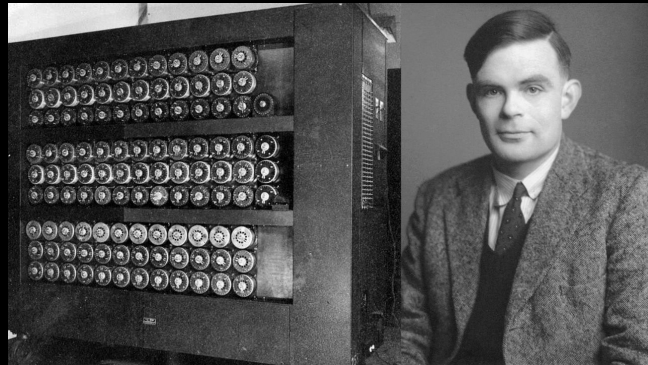
Five years later, Alan Turing opened Computing Machinery and Intelligence with the famous question, “Can machines think?” He avoided a definitive account of thought and instead proposed the imitation game, an operational test based on conversational behavior.

# AI

Alan Turing

1950

Computing and thinking



Artificial intelligence became a named field. Machine learning shifted attention from programming every rule to learning patterns from data. Deep learning used multilayer neural networks, growing data, and computation to learn increasingly powerful representations. Generative AI made the production of text, images, audio, code, and other media central. Large language models became one important family of generative systems. ChatGPT then placed these capabilities inside conversation. Today we have a fast-changing ecosystem that includes OpenAI, Anthropic, Google, DeepSeek, Qwen, Mistral, and many others. This is not a clean sequence in which one technology replaces another. These categories overlap and nest. The important change for this keynote is interface and participation. A person can now address an enormous computational and cultural apparatus in ordinary language and receive a contextual response within seconds.

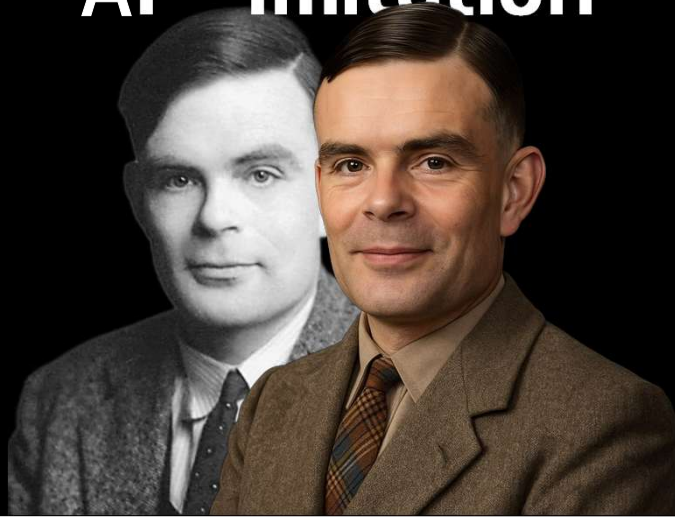
[Sources]

- <https://www.theatlantic.com/magazine/archive/1945/07/as-we-may-think/303881/>

- <https://academic.oup.com/mind/article/LIX/236/433/986238>

[/Sources]

# AI = Imitation

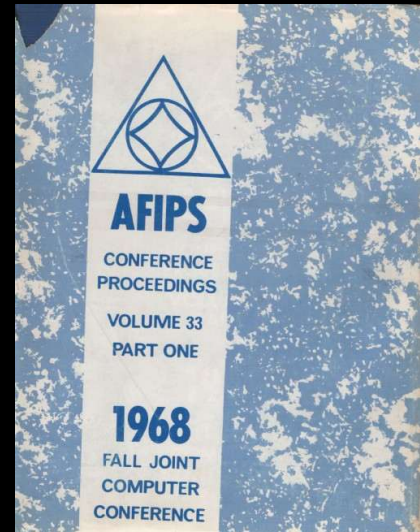


## Immersion

Ivan Sutherland

1968

A Head-Mounted 3D Display



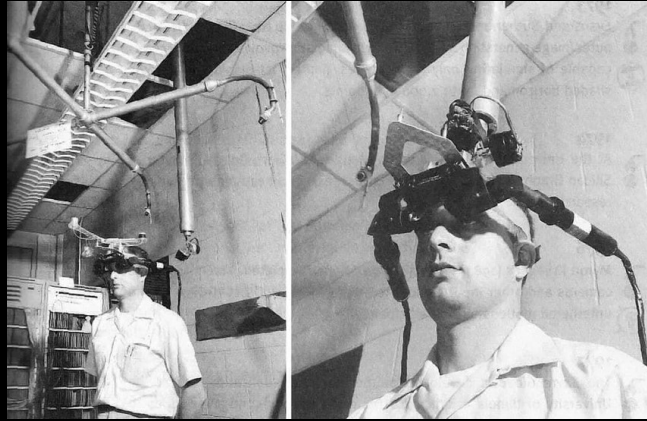
Ivan Sutherland is widely regarded as one of the founders of computer graphics and virtual reality. In 1968, together with Bob Sproull, he developed the first computer-generated head-mounted display system, later nicknamed the **Sword of Damocles** because it was suspended from the ceiling.

## Immersion

Ivan Sutherland

1968

Virtual Reality



This pioneering system tracked the user's head movements and updated a three-dimensional graphical scene accordingly, introducing a core principle of modern virtual reality: viewpoint-dependent immersive visualization. Earlier, Sutherland's Sketchpad system had already demonstrated interactive graphical computing. His work established the technological foundations for virtual reality, augmented reality, computer graphics, and immersive visual computing. **References**

Sutherland, I. E. (1963). *Sketchpad: A Man-Machine Graphical Communication System*. MIT.

Sutherland, I. E. (1965). *The Ultimate Display*. Proceedings of IFIP Congress.

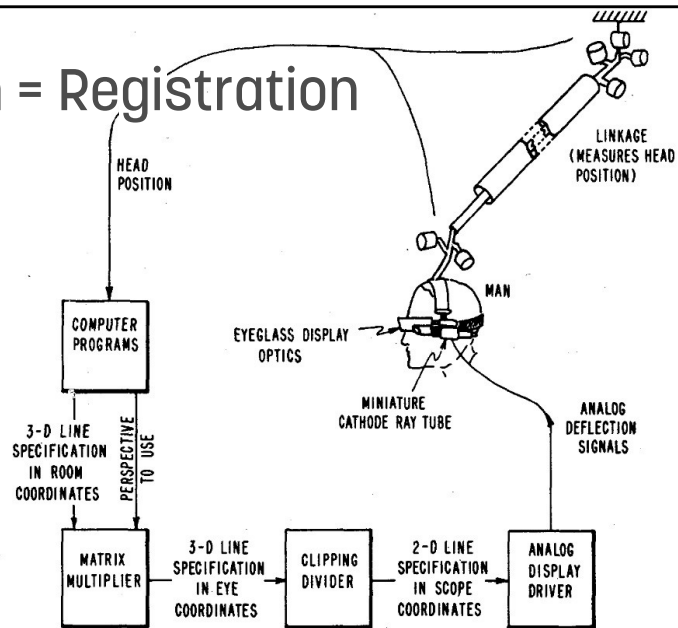
Sutherland, I. E. (1968). *A Head-Mounted Three Dimensional Display*. With Bob Sproull. AFIPS Fall Joint Computer Conference.

Wikipedia: [https://en.wikipedia.org/wiki/Ivan\\_Sutherland](https://en.wikipedia.org/wiki/Ivan_Sutherland)

Wikipedia:

[https://en.wikipedia.org/wiki/The\\_Sword\\_of\\_Damocles\\_\(virtual\\_reality\)](https://en.wikipedia.org/wiki/The_Sword_of_Damocles_(virtual_reality))

# Immersion = Registration



# Types of cognition

| Process               | Primary locus                                    |
|-----------------------|--------------------------------------------------|
| Internal cognition    | In your mind                                     |
| External cognition    | In mind and through representations              |
| Distributed cognition | Across people and artifacts                      |
| Embodied cognition    | In and through the body                          |
| Enactive cognition    | In action with the world                         |
| Immersive cognition   | Cognition in virtuality (traditional definition) |

# Literature overview of cognition

| Concept                                           | Main authors and representative publications                                                                                                                                                                                                                |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Internal cognition                                | Classical cognitive psychology; <a href="#">Ulric Neisser</a> (1967), Cognitive Psychology; <a href="#">Allen Newell</a> & <a href="#">Herbert Simon</a> (1972), Human Problem Solving                                                                      |
| External cognition                                | <a href="#">Donald Norman</a> (1993), Things That Make Us Smart; <a href="#">John M. Carroll</a> and HCI tradition; <a href="#">David Kirsh</a> (1995, 2010s) on cognitive offloading and complementary action                                              |
| Distributed cognition                             | <a href="#">Edwin Hutchins</a> (1995), Cognition in the Wild                                                                                                                                                                                                |
| Embodied cognition                                | <a href="#">George Lakoff</a> & <a href="#">Mark Johnson</a> (1980); <a href="#">Francisco Varela</a> , <a href="#">Evan Thompson</a> , & <a href="#">Eleanor Rosch</a> (1991), The Embodied Mind; <a href="#">Paul Dourish</a> (2001), Where the Action Is |
| Enactive cognition                                | <a href="#">Francisco Varela</a> , <a href="#">Evan Thompson</a> , & <a href="#">Eleanor Rosch</a> (1991), The Embodied Mind; <a href="#">Evan Thompson</a> (2007), Mind in Life                                                                            |
| Extended mind                                     | <a href="#">Andy Clark</a> & <a href="#">David Chalmers</a> (1998), The Extended Mind                                                                                                                                                                       |
| Cognitive offloading                              | <a href="#">David Kirsh</a> (1995, 2010s); <a href="#">Andy Clark</a> (1997, 2008)                                                                                                                                                                          |
| Immersive cognition (XR interpretation)           | <a href="#">Mel Slater</a> , Slater, M. (2018). Immersion and the illusion of presence in virtual reality. <a href="#">Ivan Sutherland</a> A Head-Mounted Three-Dimensional Display (1968). <a href="#">Jeremy Bailenson</a> Experience on Demand (2018).   |
| Automation                                        | <a href="#">Lisanne Bainbridge</a> (1983), Ironies of Automation; later human-automation interaction literature                                                                                                                                             |
| Human-AI collaboration / mixed-initiative systems | <a href="#">Eric Horvitz</a> (1999), mixed-initiative interaction; <a href="#">Ben Shneiderman</a> (2020), Human-Centered AI                                                                                                                                |

# Immersive Cognition in XR



Mel Slater (1990s-present)

- VR gives psychological illusion of presence
- People often respond to VR as real
- Cognition changes through convincing experiences

# Immersive Cognition in XR



- XR influences memory, attitudes, and behavior
- Embodiment and perspective-taking
- VR Experiences can be transformative
  - Psychologically
  - Behaviorally

Jeremy Bailenson (2000s-present)

# Immersive Cognition

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## Sensemaking is association

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## AI is imitation

---

## Immersion is registration

---

# Immersive Cognition

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Sensemaking is ~~association~~ *discernment*

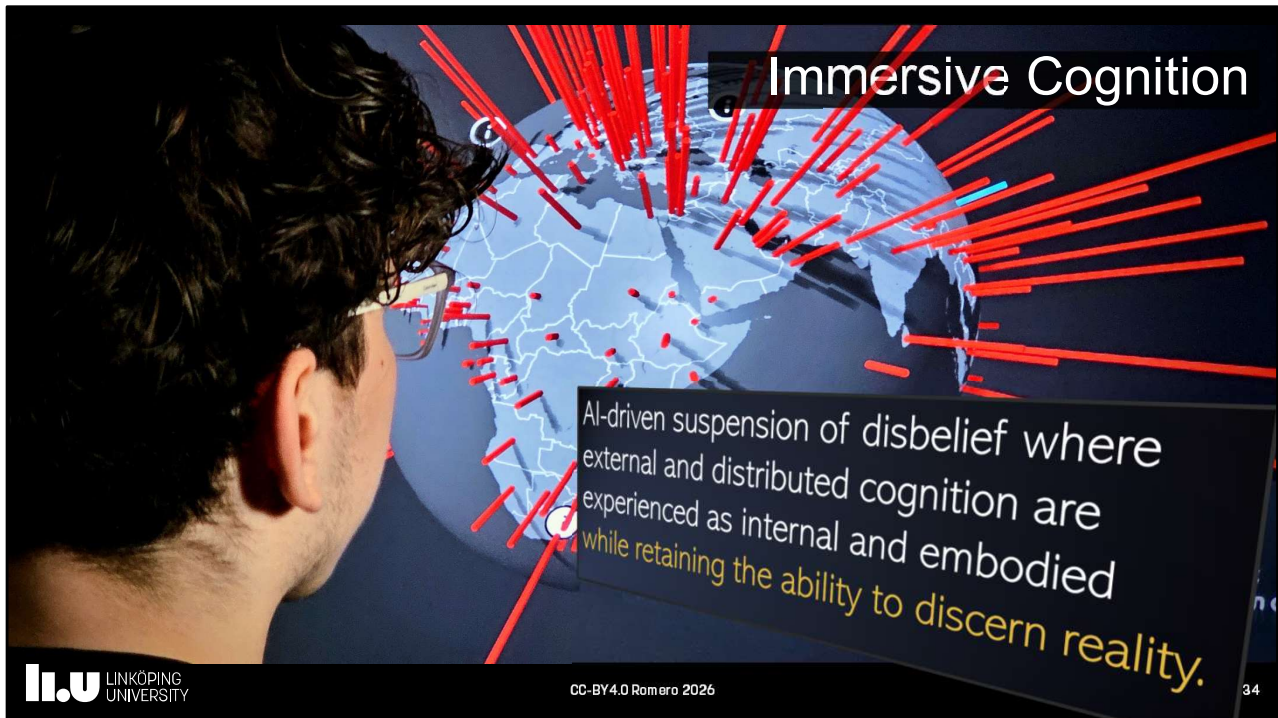
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AI is ~~imitation~~ *alien*

---

Immersion is ~~registration~~ *suspension of disbelief*

---



I began with a small puzzle and a question about cheating. The deeper question was where we draw the boundary of thinking. Humans have always thought internally, but civilization accelerated when thought could be externalized, preserved, inspected, shared, and accumulated. Bush imagined a personal machine for navigating associative trails through the growing record. Turing asked how machine intelligence might be recognized through interaction. Today, conversational and multimodal systems bring those trajectories together. They place computational capability and externalized knowledge inside increasingly immediate dialogue. But capability is not the same as judgment, and automation is not the same as understanding. The design opportunity is to help people and groups think better, question more carefully, perceive relationships, retain agency, and remain accountable. That is what I mean by immersive cognition: external and distributed resources becoming integrated into a continuous process of thought. For thousands of years, humanity advanced by externalizing thought. Perhaps the next step is to internalize it at scale. Thank you. I look forward to your questions and, especially, your disagreements.

[Sources]

- <https://www.theatlantic.com/magazine/archive/1945/07/as-we-may-think/303881/>
  - <https://academic.oup.com/mind/article/LIX/236/433/986238>
  - Internal project grounding: MAW AI-Enabled AI Guide Progress Report Draft; UnitedXR Europe 2026 Abstract; Brossier et al. OpenSpace NLI paper; Alessandro Iop project materials.
- [/Sources]

# Design lessons for human-centered AI for immersive cognition

- Task before technology
- Use AI first to test and question, not merely to solve
- Calibrate trust in the AI and in yourself and discern everything
- Preserve provenance, alternatives, and uncertainty
- Test early and fail quickly
- Build a robust technical stack
- Move beyond the prompt to voice, visuals, sound, touch, space, and action
- Design for collaboration, document everything with transparency

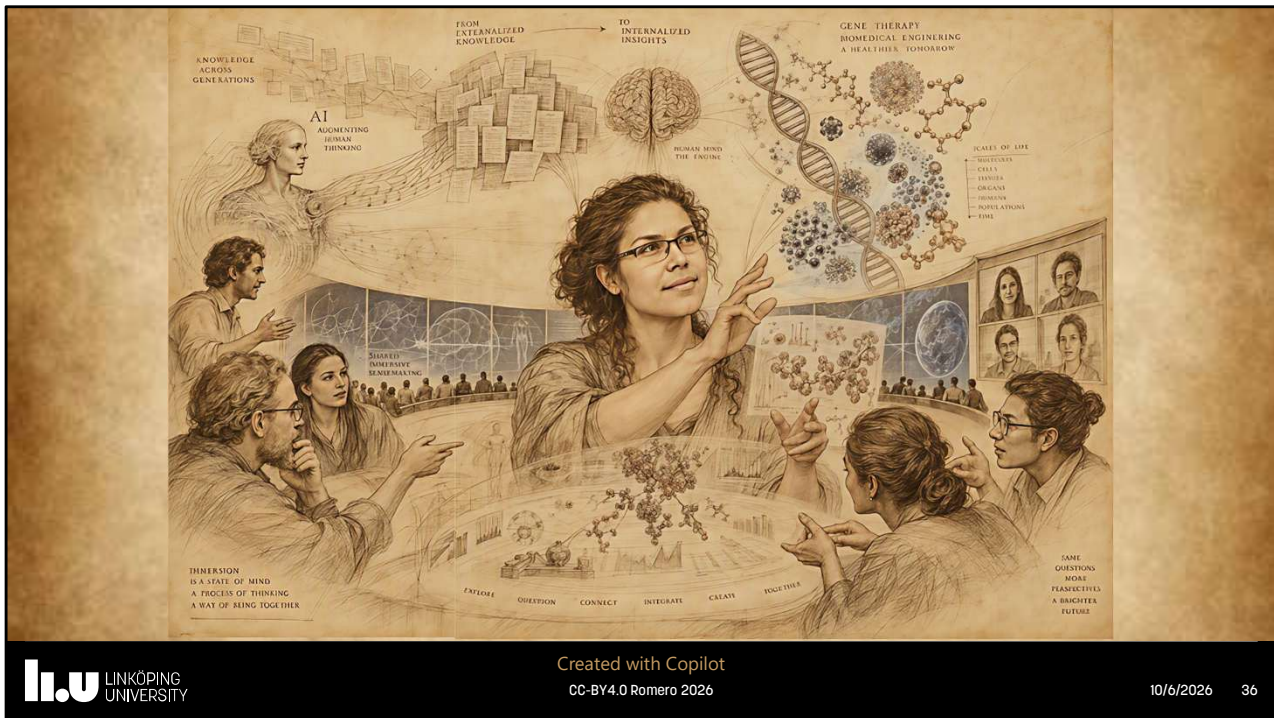
Let me end the examples with practical lessons. First, task before technology. Begin with what people are trying to understand, decide, learn, or communicate. Second, use AI first to test and question, not merely to solve. Ask it for alternatives, counterexamples, contradictions, and missing assumptions. Third, calibrate trust. The goal is neither trust nor distrust, but the right degree of reliance in the right context. We must also know when to doubt ourselves. Fourth, preserve provenance, alternatives, and uncertainty so that fluent output does not become invisible authority. Fifth, test early and fail quickly. A conceptual weakness discovered in a sketch is cheaper than one discovered in a finished dome production. Sixth, build a robust technical stack. A brilliant live demonstration can still fail because a microphone is muted or a network connection drops. Seventh, move beyond the prompt. Voice, visuals, sound, mobile devices, tangible objects, spatial interaction, and action can all support cognition. Finally, design for collaboration and document everything, including failures, repairs, and unexpected uses.

[Sources]

- <https://www.theatlantic.com/magazine/archive/1945/07/as-we-may-think/303881/>

- <https://academic.oup.com/mind/article/LIX/236/433/986238>
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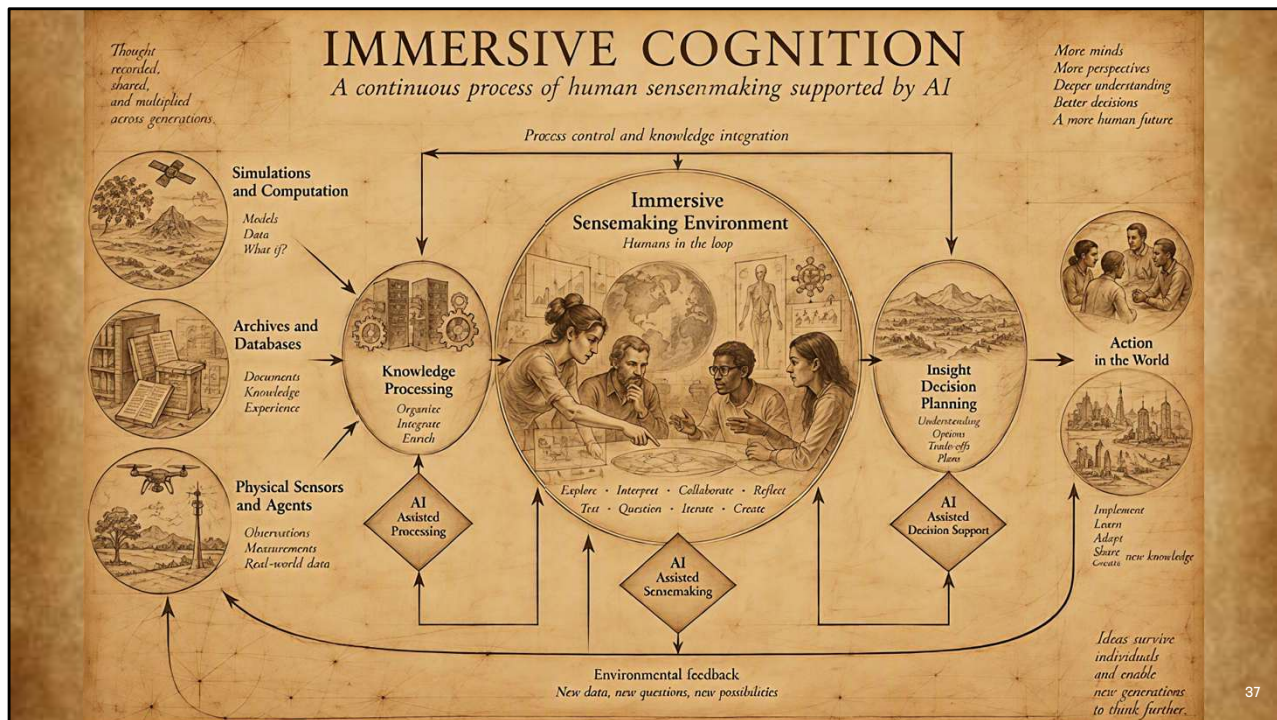


Now we need the second term in the title: immersion. In computing, immersion is often associated with a first-person perspective registered to an environment. Head tracking updates the view. Geometry responds consistently. Virtual, augmented, and mixed reality systems can make digital content appear spatially present around the user. This technological definition has been enormously productive. I am not rejecting it. I am proposing a broader layer. We also describe people as immersed in a book, absorbed by music, lost in calculation, or fully engaged in conversation. Here immersion concerns sustained cognitive and sometimes embodied engagement. It is not binary. The useful question is not simply whether a person is immersed, but how immersed, in what, with whom, and toward which purpose. A technically immersive headset can still produce boredom, confusion, or divided attention. A pencil and paper can support extraordinary depth of engagement. Technology can afford immersion by reducing friction, coordinating representations, supporting action, and connecting people. It cannot guarantee immersion because immersion also depends on task, meaning, skill, agency, and the dynamics of attention.

[Sources]

- <https://www.theatlantic.com/magazine/archive/1945/07/as-we-may-think/303881/>
  - <https://academic.oup.com/mind/article/LIX/236/433/986238>
  - Internal project grounding: MAW AI-Enabled AI Guide Progress Report Draft; UnitedXR Europe 2026 Abstract; Brossier et al. OpenSpace NLI paper; Alessandro Iop project materials.
- [/Sources]

**Prompt:** Generate a Leonardo Da Vinci style image of a biomedical engineer immersed in data sensemaking about molecules and gene therapy. She is a Latin Aamerican woman. Other people in the same room and online are working with her. She is using AI to pull vast amounts of research literature to support her conclusions. The visualizations are rendered in the environment. People are not wearing heads up displays. It is an ecosystem of intelligence, data, and visualizations. Do not include other elements. Leave blank spaces where possible. Render a high-resolution 16:9 landscape scene.



Three intellectual traditions converge in the technologies we use today. The first is **sensemaking**, the human process of constructing understanding from experience, data, and representations. The second is Vannevar Bush's Memex, which envisioned a machine for preserving, organizing, and navigating externalized knowledge through associative links rather than rigid hierarchies. The third is artificial intelligence, which enables conversational access to information and the dynamic generation of new interpretations. At the same time, immersive technologies make knowledge increasingly immediate and experiential. Together, these paths converge toward immersive cognition: a future in which externalized knowledge, intelligent systems, and human thought participate in a continuous process of sensemaking. [\[csee.umbc.edu\]](http://csee.umbc.edu/), [\[en.wikipedia.org\]](http://en.wikipedia.org)

### Sources

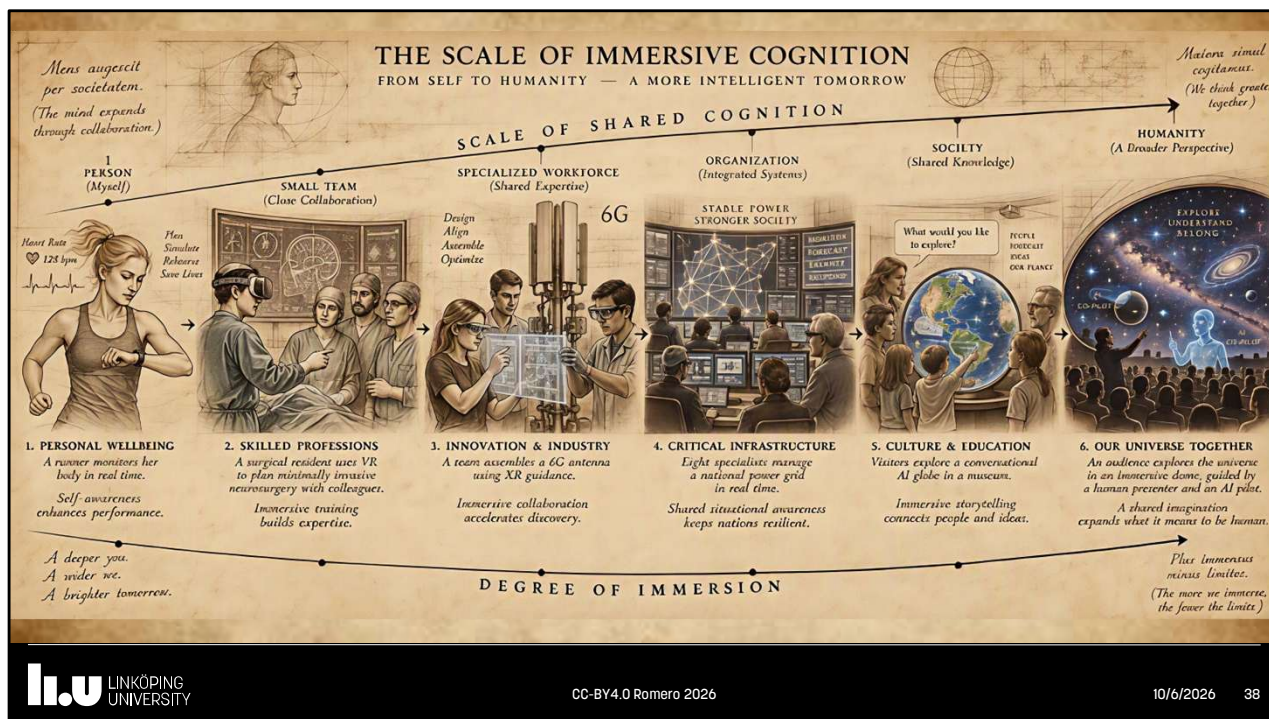
Bush, V. (1945). As We May Think. [The Atlantic](http://www.theatlantic.com)

Turing, A. M. (1950). Computing Machinery and Intelligence. [Wikipedia summary \[en.wikipedia.org\]](http://www.wikipedia.org)

Engelbart, D. C. (1962). Augmenting Human Intellect: A Conceptual Framework.

Pirolli, P., & Card, S. (2005). The Sensemaking Process and Leverage Points

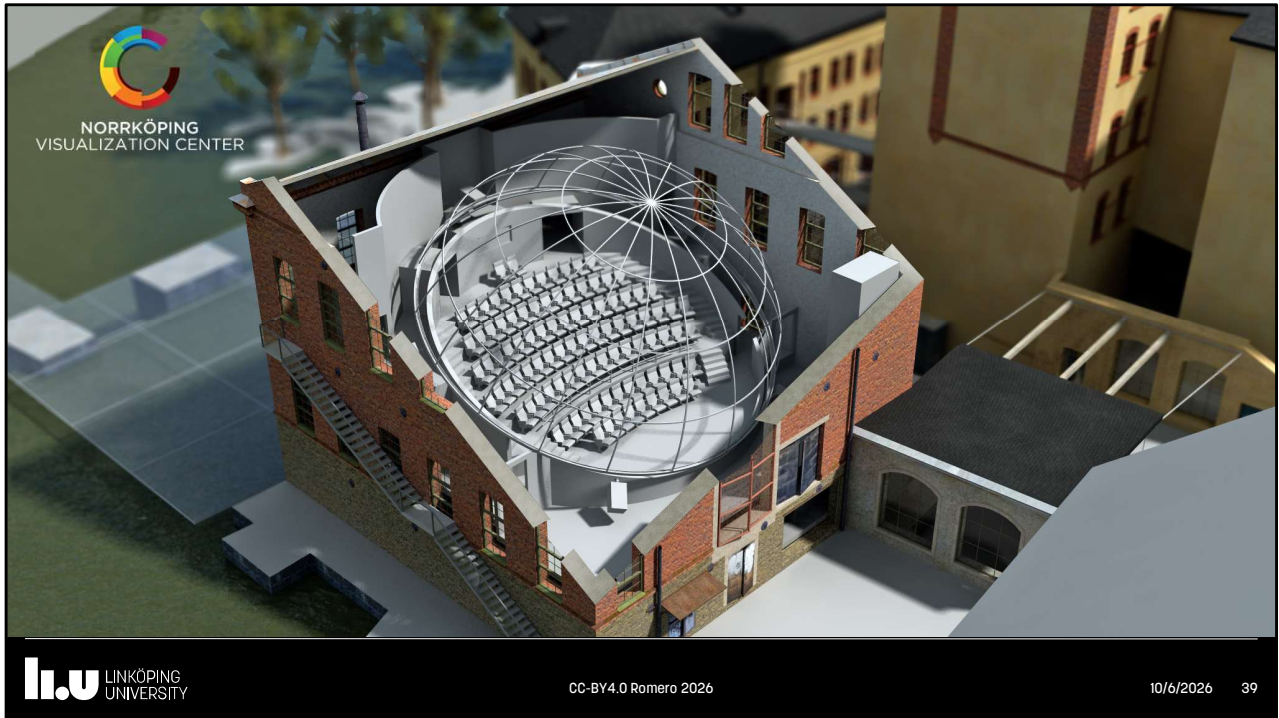
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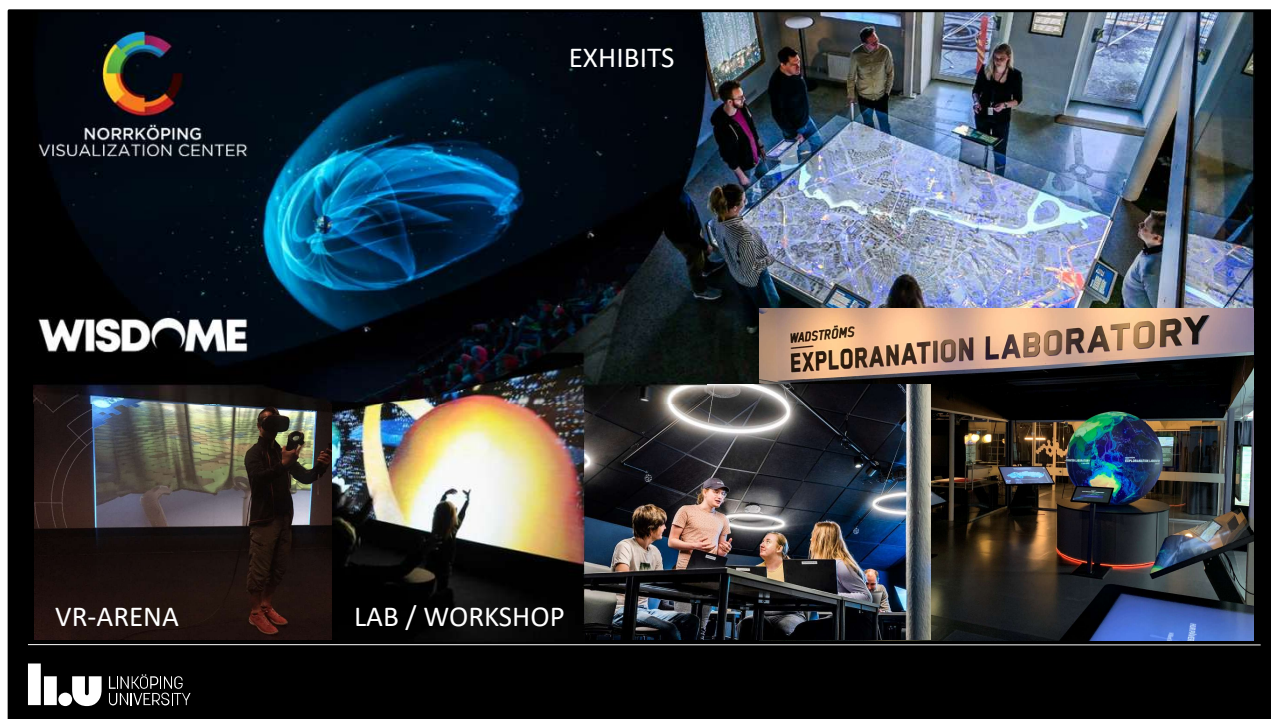


One familiar history of immersion follows technologies. Perspective images and panoramic environments organize vision around a viewer. Stereoscopy and cinema create depth, motion, and enveloping experience. Simulation binds representation to action. Head-mounted displays register a view to head movement and increasingly to hands, bodies, and physical space. CAVEs and large displays support room-scale and sometimes collaborative interaction. Mobile augmented reality overlays digital information on the world. Contemporary VR and mixed reality combine spatial graphics, sound, sensing, and interaction. Domes give groups a shared field of view at architectural scale. Networked environments can connect participants across locations. This history matters, but it tends to treat immersion as an effect produced by a device. My proposed shift is to ask what the device allows people to think, perceive, coordinate, and understand. The headset, table, dome, phone, or wall is not the unit of analysis. The unit of analysis is the cognitive process supported across a person, other people, representations, AI, and the environment. The same device may support deep immersion in one task and shallow spectacle in another.

[Sources]

- <https://www.theatlantic.com/magazine/archive/1945/07/as-we-may-think/303881/>
  - <https://academic.oup.com/mind/article/LIX/236/433/986238>
  - Internal project grounding: MAW AI-Enabled AI Guide Progress Report Draft; UnitedXR Europe 2026 Abstract; Brossier et al. OpenSpace NLI paper; Alessandro Iop project materials.
- [/Sources]









## Visualizing learner skills in minimally invasive neurosurgical simulations



The image shows three people (two men and one woman) sitting at a table in a room. On the table are two laptops, a white skull model, and some papers. In the background, a large monitor displays a simulation interface with a 3D model of a brain and a list of items. To the left of the table is a poster titled "ITN - MIT" with a colorful, abstract graphic. To the right is a green banner with the text "GUIDE IN THE IP JUNGLE" and a list of categories: Patents, Medical Devices, Designs, Litigation, Trademarks, and the website kipa.se. The bottom of the image features the Linköping University logo and contact information.

**li.u** LINKÖPING UNIVERSITY

Alessandro Iop, Martin Falk, Peter Steneteg, Andrii Matviienko, **Mario Romero**  
and introducing Arina Shahbazi

contact: [alessandro.iop@liu.se](mailto:alessandro.iop@liu.se)

## Visualizing learner skills in minimally invasive neurosurgical simulations



**ITN - MIT**

**GUIDE IN THE IP JUNGLE**  
Patents  
Medical Devices  
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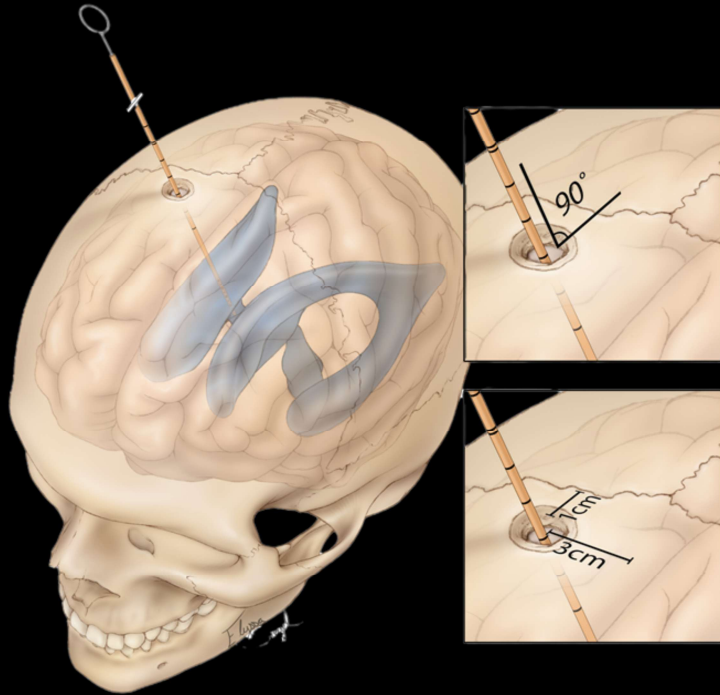
**li.u LINKÖPING UNIVERSITY**

**Alessandro Iop, Martin Falk, Peter Steneteg, Andrii Matviienko, Mario Romero**  
and introducing Arina Shahbazi

contact: [alessandro.iop@liu.se](mailto:alessandro.iop@liu.se)

# Neurosurgery

## External Ventricular Drain



EVD placement by Neurological Atlas Elyssa Siegel.

## Understand and enhance human learning of surgical skills

- Enhance through technology
  - Motion capture
  - Sensors
  - Computer vision
  - Data visualization
  - Immersive interaction
  - AI conversational agents
  - Visual-verbal analysis



Phantom

3D Print

Skull  
Brain



Phantom

3D Print

Skull  
Brain



Observe &  
analyze  
real practices

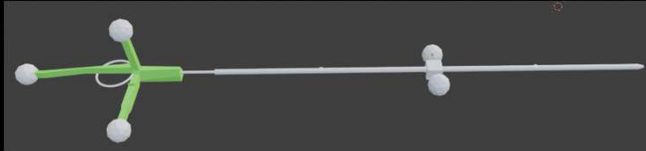






# Measure performance

- Simulation environment
- Data collected
- Data processing, cleaning
- Computations and analyses
- Statistical test results



- Brief summary of our simulation environment: phantom, MoCap, conditions, population
- Collected data was labelled and gaps were filled based on calibrated rigid bodies
- Estimating the position of the catheter tip was more error-prone
- After processing in the MoCap sw, there may still be noise and shakiness, and the recordings include irrelevant movements before and after the simulation → smoothing and cropping
- PARENT metrics were computed and analysed
- Test results will be shown in a later slide

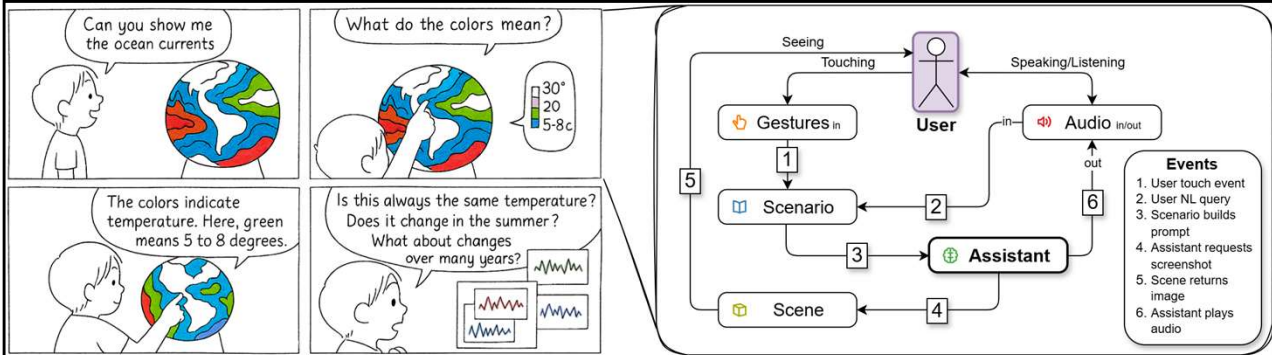




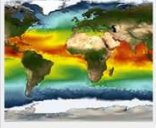
The research project “AI-enabled conversation in Science Centers and Museums” forms the foundation for the application project \*TellUs – The Talking Planet\* within WISDOME. \*TellUs\* is an example of a Wisdome project in which we explore and develop new, innovative ways to spark an interest in STEM among children and young people.

How do we do this? We base our work on research-driven pedagogical methods to create engaging, interactive, and tailored learning experiences, combining them with advanced technology such as visualizations and generative AI. This development process is also grounded in scientific research and evidence-based practice.

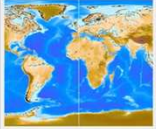
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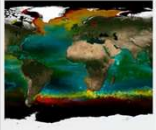
Dataset Selection



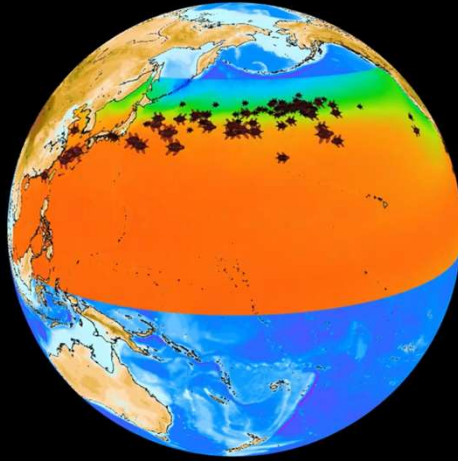
Sea Surface Temperature



Loggerhead Sea Turtles



Phytoplankton



59

How can I help you?...

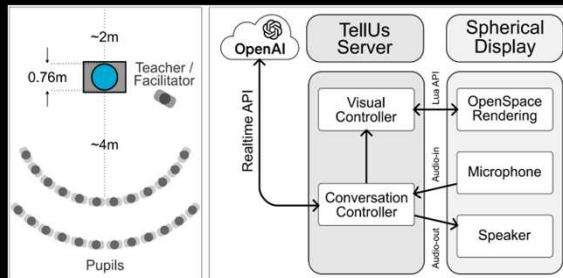
Send Record



- 2M Globe
- 300 LED panels
- Live ChatGPT
- 100 different datasets
- 700 people in the room

TellUs is on the road!

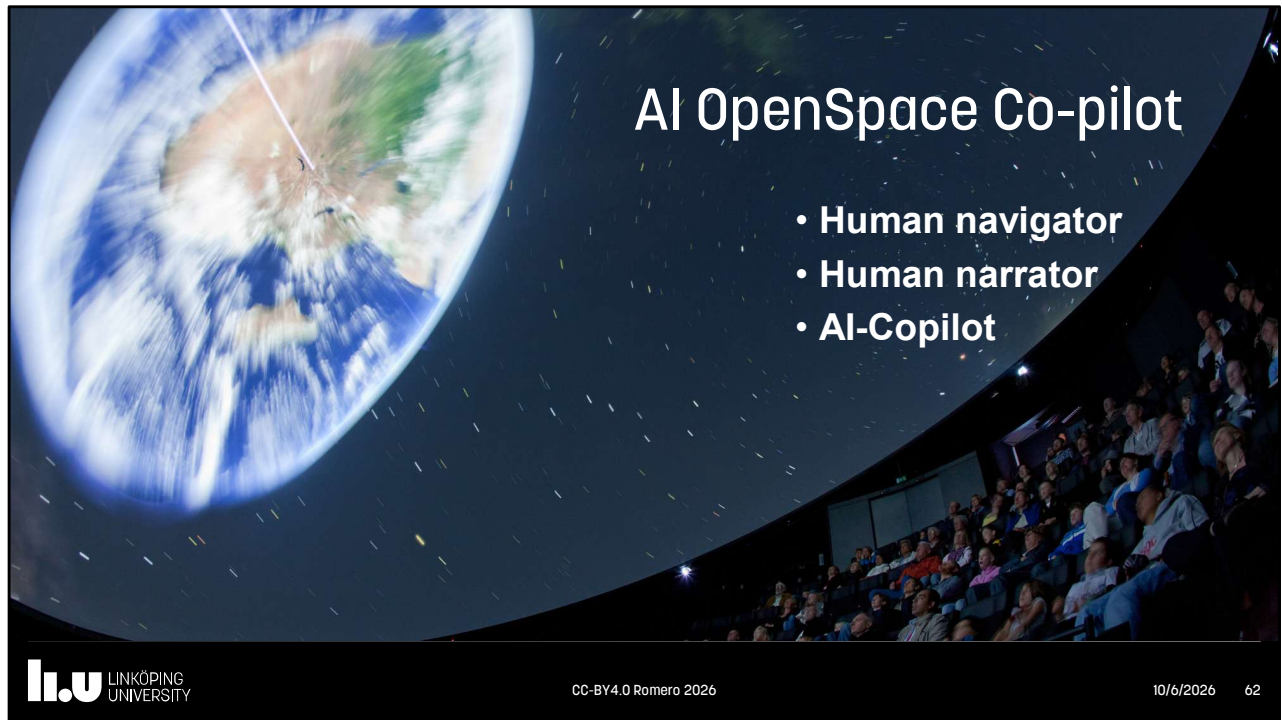
## Deployments in schools



- Conducted at Söderporten School.
- Met with approximately 100 children (six Grade 1 classes and four Grade 2 classes) and six teachers.
- We have quantitative targets in project C: 1,200 children, 500 learning sessions, and professional development for 300 teachers.
- This autumn, our first program offering will go live. We then plan to visit schools in Norrköping and the Östergötland region approximately twice a week.
- Part of our project involves researching learning outcomes related to TellUs. The research team includes Isabelle, Andreas Göransson, Konrad Schönbörn, and myself.
- During the pilot, we are testing a method in which children visually communicate their mental models and imagined worlds.
- We have given them a pre-visit task: draw what the Earth looks like before the visit. After the visit, they are asked to draw the Earth for someone who has never seen it. We want to see whether they draw differently after the

visit, whether they focus on specific aspects, and whether elements from the data we presented appear in their drawings.

- They also answer questions about their experience of the visit, where they indicate their responses by circling different emojis.



# AI OpenSpace Co-pilot

- Human navigator
- Human narrator
- AI-Copilot

**li.u** LINKÖPING UNIVERSITY

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# WISDOME

WALLENBERG IMMERSIVE SCIENCE COMMUNICATION DOME

- Visualiseringscenter C, Norrköping
- Malmö Museum, Malmö
- Universeum, Göteborg
- Tekniska, Stockholm
- Curiosum, Umeå



*Knut och Alice  
Wallenbergs  
Stiftelse*

CURIO S U M

Malmö Museer

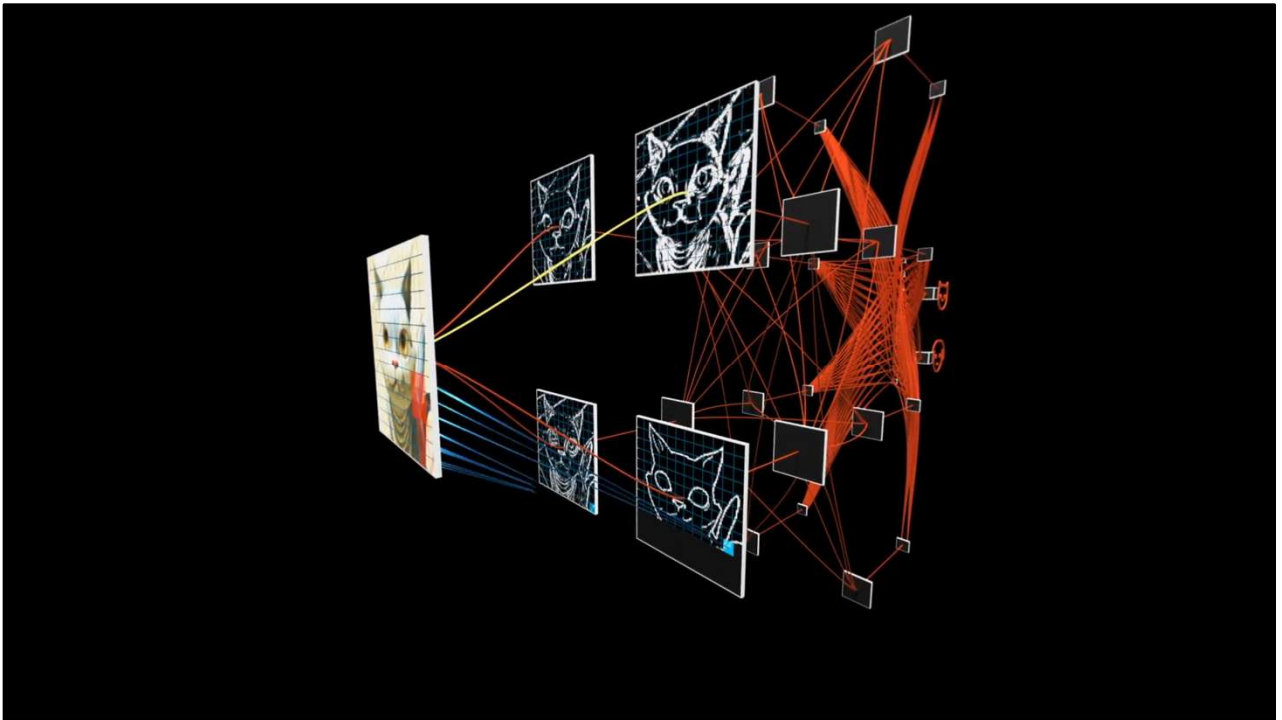


TEKNISKA









3D dome movie visualizing how deep neural networks work.



**Alexander Bock**  
Assistant Professor



**Emma Broman**  
Research Engineer



**Malin Ejdbo**  
Research Engineer



**Andreas Engberg**



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**Josefin Dellner**  
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**Adrian Elmi**  
Neurosurgeon Uppsala, Capio



**Erik Edström**  
Neurosurgeon Karolinska I. & Capio



**Andrii Matvienko**  
Assist. Professor KTH

## Recent Select Publications [\(Google Scholar\)](#)

### 2026

- Children's Expectations, Engagement, and Evaluation of an LLM-Enabled Spherical Visualization Platform in the Classroom
- State of the Art of LLM-Enabled Interaction with Visualization
- Piloting Planetarium Visualizations with LLMs During Live Events in Science Centers
- The Right Voice for the Right Task: Exploratory Insights on Persona and Voice Embodiment in Conversational AI for Industrial Training
- The Use of Augmented Reality for Navigation in Minimally Invasive Abdominal and Thoracic Soft-Tissue Surgery: A Systematic Review

### 2025

- A Scoping Review and Expert Recommendations for Immersive Solutions Towards Predictive Maintenance
- PerfCam: Digital Twinning for Production Lines Using 3D Gaussian Splatting and Vision Models
- Towards Enhancing Industrial Training Through Conversational AI
- Enhancing Manufacturing Training Through Augmented Situated Visualization
- Situated Visualization Towards Manufacturing Maintenance Training: Scoping Review, Design and User Study
- Sound-Assisted Climbing: Exploring Bouldering Performance Through Auditory Guidance
- Interactive Tangible and Intangible Heritage Applications Built on Real-Time Graphics Systems
- A Study of Multimodal Pen + Gaze Interaction Techniques for Shape Point Translation in Extended Reality
- Eyes on Target, Pen on Table: Precise VR Interaction with Gaze and Pen
- Addressing Challenges in Recruiting Proficient Braille Typists for User Studies
- Modelling Medical Students with Engineering Students: Addressing the Knowledge Gap in User Studies

### 2024

- Extended Reality in Cranial and Spinal Neurosurgery: A Bibliometric Analysis
- Assistive Augmented Reality for Adults on the Autism Spectrum with Intellectual Disability
- An Extended Reality Platform for Inclusion of Adults on the Autism Spectrum
- Augmented Reality Navigation in External Ventricular Drain Insertion: A Systematic Review and Meta-Analysis
- Exploring the Influence of Object Shapes and Colors on Depth Perception in Virtual Reality for Minimally Invasive Neurosurgical Training
- Comparative Analysis of Spatiotemporal Playback Manipulation on Virtual Reality Training for External Ventricular Drainage

### 2023

- Improving the Learning of Mechanics Through Augmented Reality
- Adopting Extended Reality? A Systematic Review of Manufacturing Training and Teaching Applications
- On Extended Reality Objective Performance Metrics for Neurosurgical Training
- A Student-Centered Learning Analytics Dashboard Towards Course Goal Achievement in STEM Education

### 2022

- An Experimental Study of the Impact of Virtual Reality Training on Manufacturing Operators on Industrial Robotic Tasks
- Extended Reality in Neurosurgical Education: A Systematic Review

### 2021

- Visualizing Bacteria-Carrying Particles in the Operating Room: Exposing Invisible Risks

# How should we use AI for immersive cognition?



**menti.com**  
**7852 8274**

# Connect



A LinkedIn profile card for Mario Romero. The background image shows a city street with historic buildings and a sunset. The card includes a blue LinkedIn icon, a circular profile picture of Mario Romero, a QR code, and his professional details.

**in**

**Mario Romero**  He/Him

Senior Associate Professor LiU, Lead Immersive Visualization Unit, National Technical Manager InfraVis, Scientific Lead Immersive & Visual Sweden, Director of C-Research. Focus: multimodal immersive visualization and AI.

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 Linköping University

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